**Soln:** Let $p, q, r$ have the same truth value. $\therefore$ By defn. of negation, $(\neg p, \neg q, \neg r)$ have same truth value w.r.t. each other but it is the opposite of $p, q, r$'s truth value.

By defn. of OR, $(p \vee q \vee r)$ have the same truth value as $p, q, r$.
By defn. of OR, $(\neg p \vee \neg q \vee \neg r)$ have the same truth value as $\neg p, \neg q, \neg r$.
$\therefore$ By defn. of AND, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ must be False.

Let $p, q$ have opposite truth values. By defn. of OR, $(p \vee q \vee r)$ must be True.
By defn. of negation, $\neg p, \neg q$ have same truth value.
By defn. of negation, $\neg p, \neg q$ have opposite truth value.
By defn. of OR, $(\neg p \vee \neg q \vee \neg r)$ must be True.
$\therefore$ By defn. of AND, $\boxed{(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)}$ must be True.

Similarly for the cases where ① $p, r$ have opposite truth values ② $q, r$ have opposite truth values, $\boxed{(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)}$ must be True.

44) If $p_1, p_2, \ldots, p_n$ are $n$ propositions explain why $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^{n} (\neg p_i \vee \neg p_j)$ is true if and only if atmost one of $p_1, p_2, \ldots, p_n$ is true.

**Soln:** **Direction 1:** $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^{n} (\neg p_i \vee \neg p_j)$ is True $\Rightarrow$ atmost one of $p_1, p_2, \ldots, p_n$ is true.

**Contrapositive:** Atleast two of $p_1, p_2, \ldots, p_n$ is true $\Rightarrow$ $\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^{n} (\neg p_i \vee \neg p_j)$ is False

**Proof:** Let, wlog $p_i, p_j$ be true, where $1 \leq i < j \leq n$. By defn. of negation, $\neg p_i, \neg p_j$ is False. By defn. of OR logic, $(\neg p_i \vee \neg p_j)$ is False.
By defn of AND, $\boxed{\bigwedge_{j=i+1}^{n} (\neg p_i \vee \neg p_j) \text{ is False}}$
By defn of AND, $\boxed{\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^{n} (\neg p_i \vee \neg p_j) \text{ is False}}$

An implication and its contrapositive are logically equivalent.
$\therefore$ Direction 1 is proved.